

Simmon Yang

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SUMMARY

Electrical Engineering Technology graduate (with Honours) and G3-certified Gas Technician (G2 in progress) with hands-on experience across gas and HVAC installation, electrical and low-voltage infrastructure, embedded systems, and technical troubleshooting. Combines field execution (gas piping, venting, ductwork, wiring, testing, code-compliant installations) with software and automation skills (C/C++, Python, PLC, TypeScript) to deliver reliable, practical systems. Seeking technical roles in gas, mechanical, electrical, and controls environments.

TECHNICAL SKILLS

- **Electrical & Hardware:** Residential & Commercial Wiring, Low-Voltage Systems, DC/AC Circuits, Digital Circuits, Testing & Troubleshooting, Microcontrollers (Arduino, Raspberry Pi), PCB Design (KiCad), Security Systems, Network Cabling (Cat5e/Cat6/Cat6A), Soldering
- **Mechanical & Gas:** Gas Piping Installation, Venting Systems, Sheet Metal Ductwork, Supply & Return Air Distribution, PVC Intake/Exhaust Piping, Tankless Water Heaters, HVAC Installation & Service, Copper Cutting & Brazing
- **Software & Tools:** C, C++, Python, TypeScript, Java, PLC Programming, MySQL, SQLite, PHP, WordPress, React Native, Docker, Git, AutoCAD Electrical, Multisim, Claude (AI-Assisted Development & Workflow Automation)
- **Standards & Safety:** Electrical Code Comprehension, Gas Code Comprehension (G3), TIA/EIA Standards, Lockout/Tagout, WHMIS, Working at Heights

EDUCATION

Electrical Engineering Technology Diploma (with Honours) Sep 2022 – Apr 2026
Centennial College *Scarborough, ON*

- Relevant coursework: Electrical Theory, Drawing Interpretation & Installation, Monitoring Systems, Networking, PLC Programming, Power Generation, Microcontrollers, Signals & Communication
- GPA: 4.1

Gas Technician 2 (G2) – In Progress 2026 – Present
Himark College *Ontario*

- Currently completing G2 certification coursework; TSSA certification exam scheduled for September 11, 2026.
- Coursework covers advanced gas appliance installation, venting design, combustion analysis, and code compliance beyond the G3 scope.

Gas Technician 3 (G3) Certificate 2025
Stanford College *Toronto, ON*

- Completed G3 certification training covering gas properties, piping systems, venting fundamentals, appliance installation, and applicable codes and regulations.

WORK EXPERIENCE

Gas Technician Co-op Jun 2026 – Aug 2026
Considerate Tech *Greater Toronto Area, ON*

- Installed gas piping systems in new residential construction, including cutting, threading, joining, and pressure-testing runs to appliance connection points.
- Installed venting systems for gas-fired appliances, verifying correct sizing, slope, clearances, and termination requirements.
- Fabricated and installed sheet metal ductwork, including supply and return air trunks, branch runs, and register boxes from mechanical drawings.

- Installed PVC intake and exhaust piping for high-efficiency appliances, ensuring proper support, sealing, and condensate routing.
- Installed and connected tankless water heaters, coordinating gas, venting, and water line terminations for commissioning.
- Worked to code and manufacturer specifications on active construction sites while maintaining site safety practices.

Electrician Assistant (As-Needed)

Sep 2024 – Apr 2026

Independent / On-call

Greater Toronto Area, ON

- Called in periodically to assist a licensed electrician on residential and commercial jobs during high-demand periods.
- Pulled, cut, routed, and terminated wiring/conduit for lighting, receptacles, and power distribution during rough-in and finishing phases.
- Installed switches, outlets, fixtures, and breakers to electrical code requirements, with correct grounding and secure device connections.
- Supported panel upgrades and device commissioning by verifying continuity, voltage, and insulation resistance before energization.

Low Voltage Technician

Jul 2025 – Sep 2025

OneNid Inc

Greater Toronto Area, ON

- Installed and terminated Cat5e/Cat6/Cat6A cabling for IP cameras, access points, and intercom systems to TIA/EIA standards.
- Mounted, aligned, and calibrated IP/analog camera systems to improve site coverage and surveillance reliability.
- Installed and configured DVR/NVR platforms, including network connectivity and remote access setup for client monitoring.
- Integrated access control hardware (card readers, maglocks, motion sensors) with low-voltage wiring and control panels.
- Executed structured cabling from floor plans and wiring diagrams; labeled and organized runs for easier maintenance and troubleshooting.
- Validated terminations and diagnosed faults using cable testers, crimping tools, and multimeters before system handoff.

HVAC Assistant

May 2024 – Dec 2024

HVAC Contractor

Greater Toronto Area, ON

- Installed, maintained, and serviced residential/commercial HVAC systems with focus on safe and reliable operation.
- Fabricated ductwork and completed copper pipe cutting, brazing, and insulation to support field installations.
- Performed system checks, filter replacement, and refrigerant pressure monitoring to support preventive maintenance.
- Used wiring diagrams, schematics, and HVAC layouts to troubleshoot installation issues and verify correct system setup.

Web Developer (Contract)

Mar 2024 – Apr 2024

Riipen

Remote

- Designed and developed a dynamic WordPress/PHP website aligned with project requirements and user workflows.
- Implemented authentication and backend data flows with MySQL integration for persistent content management.
- Built responsive front-end layouts and deployed the production site to a live hosting environment.

Food Service Worker

May 2024 – Sep 2024

Pineapple Queen

Greater Toronto Area, ON

- Delivered fast, accurate customer service while handling cash/card transactions and maintaining quality standards.
- Managed ingredient inventory and restocking to support continuous service and minimize stockouts.
- Operated food-prep equipment safely and maintained a clean, organized workstation during peak service periods.

Residential Renovation Assistant

May 2023 – Aug 2023

Construction Inc

Greater Toronto Area, ON

- Installed electrical wiring for full-home renovations, including switches, outlets, and lighting fixture connections.
- Supported multidisciplinary renovation tasks (drywall, painting, demolition) to keep projects on schedule.

Operations & Systems Assistant

Jun 2022 – Aug 2022

Microgreen Solar Corp

Markham, ON

- Coordinated inventory and shipment logistics across product lines to support on-time fulfillment.
- Led packaging operations to consistently meet daily throughput and quality targets.
- Built a WordPress/PHP inventory system that automated order tracking, stock updates, and internal reporting workflows.
- Integrated MySQL data storage to centralize inventory records and improve cross-team visibility.

Electrical Technician Assistant

Jul 2021 – Aug 2021

Microgreen Solar Corp

Markham, ON

- Repaired and tested DC-AC inverter power packs for solar systems under multiple load conditions.
- Assembled and terminated custom power/signal cables, validating shielding, polarity, and connector integrity.
- Performed component-level PCB troubleshooting with multimeters, oscilloscopes, and soldering tools.
- Diagnosed and resolved faults across power conversion, control, and protection circuitry.
- Executed load and QA tests to verify voltage regulation, inverter performance, and safety compliance.
- Documented repairs and test procedures to improve repeatability and maintenance traceability.

PROJECTS

Smart Street Lighting System

Jan 2026 – Apr 2026

Arduino Uno R3 / Embedded Systems

- Designed an Arduino Uno R3 street-lighting controller that adjusts LED brightness using ambient light, rain, and vehicle-proximity inputs, with an emergency-vehicle wave mode for priority routing.
- Implemented PWM dimming that reduced LED energy use from 21.6 to 8.64 Wh/day (60% reduction) at 2.5W total system draw.
- Produced mechanical drawings in AutoCAD and electrical schematics in KiCad; documented the design in OACETT/IEEE-style reports.

Automated Circadian Lighting System (Retrofit)

Sep 2025 – Dec 2025

Lighting Control / Energy Efficiency

- Designed a retrofit LED lighting control system that adjusts color temperature and brightness throughout the day to support circadian rhythms.
- Developed a control profile that transitions from cool/high-intensity daytime lighting to warm/low-intensity evening lighting without replacing existing pot lights.
- Evaluated power consumption and lighting performance to validate energy savings and improved visual comfort.

Adaptive LED Headlight Simulation

Sep 2025 – Dec 2025

PIC16F747 / Embedded Systems

- Designed a microcontroller-based adaptive headlight simulation using a PIC16F747 and HC-SR04 ultrasonic sensor.
- Implemented distance-based LED control to adjust illumination zones and reduce glare exposure for oncoming vehicles.
- Processed ultrasonic echo timing with internal timers and mapped measured distance to discrete output zones.
- Programmed firmware in C (MPLAB X, XC8) using polling-based sensor acquisition and control logic.
- Tested PWM dimming behavior and identified timing constraints that introduced visible LED flicker.

4x4x4 RGB LED Cube

Dec 2021 – Jun 2023

Arduino / C++ / PCB Design

- Built a multiplexed RGB LED cube architecture that individually addresses 192 LEDs with 6 Arduino pins using shift registers and Bit Angle Modulation (BAM), avoiding per-LED driver ICs.
- Designed custom KiCad PCBs for the LED matrix and produced a 3D-printed enclosure for assembly.
- Implemented C++ animation effects (sparkles, fireworks) and extended the design approach to an 8x8x8 variant.

Almanac – Markets Intelligence & Backtesting System

Jul 2026 – Present

Python / pandas / vectorbt

- Built a local-first Python system with a data layer, strategy backtester, risk module, and portfolio monitor, verified end-to-end by a 44-test pytest suite.

- Implemented a rules-based equity screener with a forward-tracked scoreboard to measure signal performance against later outcomes rather than backfitted history.
- Scoped automated trade execution out of the system by design, gating any real-money use behind a pre-registered evaluation protocol.

Tiffany's Classroom – Automated Video Production Pipeline

Jun 2026 – Present

Remotion / TypeScript / Google TTS

- Built a programmatic video pipeline that composites narration, generated assets, and animated segments entirely in code, eliminating manual editing from the production process.
- Synchronized text-to-speech narration to on-screen content by trimming audio against frame-accurate composition timing; published three full-length episodes.
- Diagnosed and corrected asset-level alpha-keying defects in animation frames, then fixed them upstream so downstream renders inherited the correction.

Tether – Automated Task Accountability Agent

Apr 2026 – Present

Python / Docker / SQLite / Oracle Cloud

- Built and deployed a Discord-based scheduling agent running 24/7 in Docker on an Oracle Cloud VM, using Google Calendar as the scheduling source of truth and a SQLite ledger for task lifecycle state.
- Kept all scheduling and priority math deterministic in Python, restricting the LLM to intent parsing only so system behaviour stays predictable and testable.
- Designed a deadline-runway cap that limits rescheduling based on a task's own deferral count rather than user-supplied text, closing a loophole that text matching could not.
- Diagnosed and fixed a state bug where a scheduled job silently reverted user actions within one cycle, tracing the reported symptom to its actual mechanism.

PERFORMANCE LAB – Calisthenics Training Application

Apr 2026 – Present

React Native / Expo / TypeScript

- Developed a cross-platform mobile app featuring a 46-skill radial progression tree, a deterministic coaching engine (fatigue modelling, readiness scoring), and 5 screens with local persistence.
- Maintained a 127-test suite across 15 files covering the full service layer, with TypeScript strict typing throughout.
- Implemented a versioned data-migration system for on-device storage to preserve user data safely across schema changes between releases.
- Consolidated roughly 660 hard-coded style literals into a centralized design-token system, removing 183 dead style keys.

BT6 Bot – Computer Vision Game Automation

Nov 2025 – Present

Python / OpenCV / OCR

- Built an autonomous agent that plays a real-time strategy game using only screen capture and simulated input, with no access to game memory or APIs.
- Implemented OCR-based HUD parsing and an authoritative internal state model to track game entities independently of the rendered display.
- Developed a genetic algorithm that encodes strategies as chromosomes and evolves them across generations using run-derived fitness metrics; validated by 28 headless unit tests.

Pathfinding Visualization

Dec 2020 – Feb 2021

Python / Pygame

- Developed an interactive Python/Pygame grid visualizer supporting A*, BFS, DFS, and Dijkstra with step-by-step color-coded execution.
- Added real-time speed controls and keyboard shortcuts to reset grids and clear paths without restarting the application.
- Structured the codebase into PathFinder, UI, and settings modules with a central DISPLAY class managing shared state and events.

CERTIFICATIONS

- WHMIS Certification (2026)
- Working at Heights Certification (2024)
- G3 Gas Technician (2025)